15.10. RADIATION BACKGROUND

Through life, everyone is constantly exposed to ionizing radiation from a variety of sources. Table 15.8 gives radiation doses which are representative for a large number of countries with the exception of extreme conditions like cosmic radiation at high altitude (e.g La Paz at altitude 3900 m) and locations on monazite sand (e.g. Ramsar, Iran, Kerala, India, and Esperito Santo, Brazil). Table 15.9 gives the relative contribution of sources of radiation to the UK population. The values are representative for most countries in the northern hemisphere. Within the same country, the total background dose commonly varies within a factor of 3–5 between lowest and highest areas. Brick and concrete houses contain varying amounts of uranium and/or radium, depending on the source of the building material. Because of radioactive soil, all foods contain some natural radioactivity. An approximate average value in northern Europe is <0.3 Bq/kg from U-decay products in food or beverage, while vegetables in the monazite areas in India (e.g. roots) have been found to have >10 Bq/kg.

Presently great concern is directed towards people living in dwellings with high radon concentration ($\geq$400 Bq/m^3) because the radon and daughters are assumed to

Table 15.8 Effective dose (normal → elevated value) in mSv/y (UNSCEAR 1993) or average annual dose values (Nias 1990)

Source	Value
Cosmic rays	0.3 → 2.0 (LaPaz) mSv/y
Terrestrial, external	0.3 → 4.3 (Kerala) mSv/y
Intake	0.052 (ingestion), 0.01 (inhalation) mSv/y
Radionuclides in body[†]	0.23 → 0.6 mSv/y
Rn in water supplies and air:	see §13.6
Rn in body	1.3 → 10 mSv/y
Buildings[‡]: wood/gypsum	0.18 mSv/y (K 150, Ra 20, Th 5 Bq/kg)
" : typical masonry	0.7 " (K 500, Ra 50, Th 50 Bq/kg)
" : alum shale concrete	5.9 " (K 770, Ra 1300, Th 67 Bq/kg)
Nuclear power plant: to critical group (Sweden)	<0.1 mSv/y
X-ray investigation:	5–8 mGy
barium meal (intestine)	
intravenous urography	4–5 "
abdomen and lung	~1 "
mean gonadal dose (abdomen),	<0.2 "
genetically significant dose	0.1–0.5 "
dental	0.02 "
Total average	2–4 mSv/y

[†]Except Rn.
[‡]Staying indoors at 8760 h/y; masonry building in temperate climate.

Table 15.9 Relative contribution of sources of radiation to the population (UK National Radiation Protection Board, 1989); annual average dose is 2.2 mSv

Natural sources	87
radon	47%
food and drink	12%
γ's from ground and buildings	14%
γ's from cosmic rays	10%
Artificial sources	13%
medical	12%
fall-out	0.4%
work	0.2%
occupational[†]	<0.1%
others	0.4%

[†]Sellafield reprocessing plant, average 5 mSv/y

make the largest contribution to the present radiation background. Typical indoor values are 50—400 Bq/m^3 air; we discuss this further in §15.13.5.

15.11. SOMATIC EFFECTS OF LOW RADIATION DOSES

The effect of low radiation doses such as the natural radiation background or from the nuclear power fuel cycle (and possible accidents) is controversial because the evidence is inconclusive.

15.11.1. Epidemiological results

Of the large number of epidemiological investigations on the effects of low-LET low-level radiations, we will only mention a few representative ones.

Figure 15.9 shows the malignant mortality rate of the US white population accumulated over 18 years (1950—1967) in 46 states as a function of the natural radiation background. Each point represents an average of about 100 000 deaths and is thus significant. The average background value for the whole US population is 1.3mSv/y and the average annual mortality rate in cancer (horizontal line) corresponds to 0.15%. The only significant trend seems to be a decrease in cancer deaths with increasing background up to about 2.5 mSv/y. Although the background radiation in the 7 highest states is 2.1 mSv/y as compared to 1.3 mSv/y for the US average, the frequency of all malignancies is lower (126 per 100 000) in the former than in the latter (150 per 100 000) case. The frequency of leukemia, breast cancer, thyroid cancer, and all child malignancies (age 0—9 y) is slightly lower or the same. Although attempts have been made to correlate the cancer decrease rate with other factors, such as living